Can We Trust Marketing ROI? A Synthetic-Data Study of Confounding and Experimental Calibration

Hashan Fernando*

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Summary

Marketing Mix Modeling (MMM) is widely used for media budget allocation. However, its reliance on observational data leaves it vulnerable to severe confounding from confounding variables such as organic demand, ad-targeting mechanisms, and macroeconomic trends. Therefore, it is questionable to trust observational MMM outcomes entirely. In this study, we check whether an observational MMM can be trusted to recover causal ROI and, if not, what it takes to make it trustworthy. However, we do not propose new estimation methodology; the Bayesian MMM, the backdoor-adjustment framework, and the calibration-via-prior technique are all drawn directly from existing work [1, 2, 3]. Our contribution is empirical demonstration: using synthetic data with known ground truth, we show concretely when these established methods succeed, when they fail, and what it takes to correct them.

1 Introduction

1.1 The measurement problem in marketing

Marketing firms spend billions of dollars every year on a variety of marketing channels, and determining the actual return on investment (ROI) for any given channel is an extremely difficult task. The ROI for each channel must be determined by marketers in order to optimize their marketing funds. Simply, an ROI represents the actual revenue the firm earned and the unobservable revenue it would have earned if it had not spent advertising funds [4]. Because the counterfactual cannot be directly measured, the industry uses two common methodological approaches: Marketing Mix Modeling (MMM) and experimental lift tests.

The lift tests are executed at the user or geographic level using a randomized controlled trial (RCT) by randomly assigning subjects to treatment (exposed to ads) or control (unexposed) groups. Therefore, these lift tests eliminate confounding variables and result in unbiased identifications [5]. However, there are practical limitations to their application in real-world situations. These lift tests are expensive, time-consuming, and require a firm to accept the expense of suppressing ads for a control group.

To overcome these practical difficulties of the lift tests, marketing firms uses MMM.This MMM approach uses historical spend and sales data to assess the overall effect of all marketing campaigns at once [6]. This approach is comprehensive and does not require firms to pause their active marketing. However, MMM faces major challenges because of the dependence on the observational data. These data are confounded by many unknown factors; therefore, claims of the MMM are highly dependent on strict identification assumptions. This makes a big problem in modern marketing practices. The observational models that are broad enough to help allocate budgets are very prone to confounding bias, while the experimental models that prove causal validity are too expensive and time-consuming.

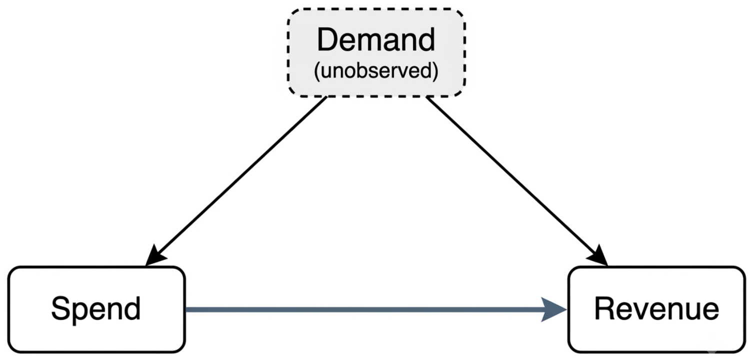


Figure 1: The confounding structure of observational marketing mix modeling. The unobserved demand variable acts as a common cause of both media spend and revenue, opening a backdoor path (Demand → Spend and Demand → Revenue) that biases naive estimates of the spend → revenue causal effect. Closing this path requires either statistical adjustment (conditioning on demand) or bypassing it via experimental calibration.

1.2 Confounding in observational MMM

Marketers set their spending based on how much demand they expect. So when spend and revenue move together, they can't tell how much of that link is the spend actually working versus both being driven by the same underlying demand. If we further explain this, marketers increase spending during peak seasons, ahead of product launches, or due to positive macroeconomic indicators. At the same time, those same factors also drive consumer demand on their own. Therefore, if a firm invests their money into marketing/ads when people are already eager to buy, a baseline regression will attribute resulting spikes in revenue to the advertising spend, making them look way more effective than they really are.

Formally, this problem can be framed within the directed acyclic graph (DAG) framework of causal inference [7] (see fig. 1). Let X_t represent media spend (the treatment), Y_t represent revenue (the outcome), and Z_t represent baseline organic demand or macroeconomic trends (the confounder). Because Z_t acts as a common cause of both the treatment and the outcome, it opens a backdoor path ($X_t \leftarrow Z_t \rightarrow Y_t$). A standard regression model that fails to block this path yields biased and inconsistent estimates of the causal effect of X_t on Y_t, picking up the non-causal association flowing through Z_t.

To recover true causal parameters from observational data, we use the backdoor adjustment [7, 3]. According to this approach, conditioning on a sufficient set of control variables is necessary to block all backdoor paths in order to evaluate the true return on investment (ROI). Considering how common unobserved confounders are in marketing data, this study explores whether we can trust marketing ROI estimates derived from observational marketing data and whether experimental calibration can correct for unobserved confounding. To explore this, we structure our investigation around three sub-questions:

1. Recovery - When the truth is known (synthetic data), do the model's posteriors cover the true ROI?
2. Confounding - How much does omitting a demand confounder bias ROI, and does adjusting for it fix the bias?
3. Calibration - Can an experimental lift estimate, injected as a prior, correct an MMM that observational data alone gets wrong?

*GitHub repository: `https://github.com/hashansl/marketing-mix-modeling-causal-inference`

2 Methodology

2.1 Overview of the experimental design

To answer the above questions, we create a synthetic dataset in which the ground-truth parameters are known by construction. This allows us to measure estimation accuracy and bias directly. Our experimental design proceeds in four stages, each addressing a distinct sub-question of the central research problem.

Stage 1 - Parameter recovery under ideal conditions: We generate a clean synthetic dataset in which media spend is assigned independently of all other variables. Therefore, conditional ignorability holds by construction, and the backdoor path identified in Figure 1 does not exist. We fit three estimators by increasing the complexity: a naive ordinary least squares (OLS) regression on raw spend, an oracle OLS regression using the true adstock and saturation transformations, and a complete Bayesian MMM, which needs to use the observational data to learn those transformations. The goal is here to recover true ROI on our clean simulated data. If we can't trust the model here, in the most favorable setting possible, we lose the controlled experiment: we can't draw causal inferences because we're not sure whether the model itself is any good.

Stage 2 - Confounding bias and backdoor adjustment: In this stage we generate a second synthetic dataset from the same structural model but introduce a latent demand confounder that affects both media spend and revenue. We fit the Bayesian MMM twice on this confounded data: first without controlling for demand (backdoor open), then with demand included as a linear control (backdoor closed). By doing this, we compare the difference in recovered ROI between these two fits and isolates the magnitude of confounding bias and the effectiveness of statistical adjustment.

Stage 3 - Experimental calibration: In the real world setting, we usually can't measure the confounder directly, so the adjustment strategy from Stage 2 won't work. To overcome this, we simulate the results of a randomized geo-lift test [8], an unbiased estimate of the ROI for one marketing channel, along with a realistic margin of error. We then plug this estimate into our Bayesian model as an informative prior on that channel's coefficient, following the calibration-via-reparameterization approach proposed by Zhang et al. [2]. Then we fit the model on the confounded data without controlling for demand, leaving the backdoor open. If the experimental prior can move the model's posterior estimate toward the true ROI even with the confounding left in, then a small amount of randomized evidence can correct for bias in a way that observational data alone cannot. This demonstrate a practical solution for estimating realistic ROI, where experiments and observational methods work together as a team.

Stage 4 - External validation on realistic data: Finally, we test our Bayesian MMM on a publicly available dataset from Meta's Robyn package: 208 weeks of media spend across five paid channels, plus a competitor-sales variable that's highly correlated with revenue. Since this dataset doesn't have known ground-truth ROI, we can't directly check whether the model recovers the right answer. Instead, we drop the competitor-sales control and see how much the ROI estimates shift.

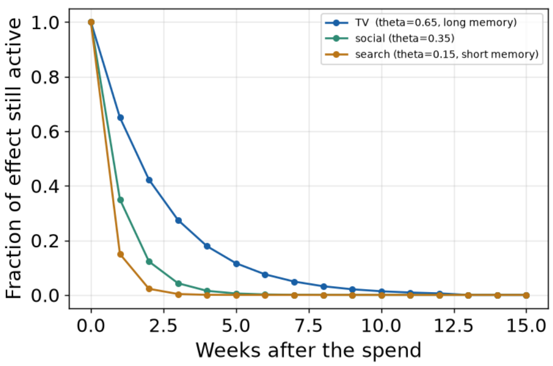
Across all four stages, we keep the Bayesian model setup the same: priors, adstock, saturation, seasonality, and sampler settings, unless we explicitly say otherwise. That way, any differences in results between stages come from the data and controls, not from us changing the model. The next subsections walk through the data-generating process, the three modeling approaches, and the calibration mechanism in full technical detail.

2.2 The data-generating process

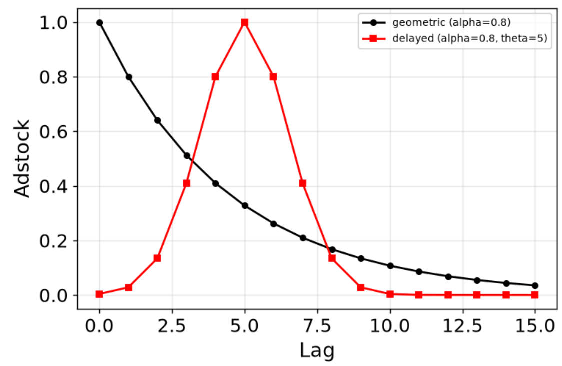
We now specify the data-generating process. Our modeling includes a mix of five distinct marketing channels: TV, Search, Social, Display, and Email. The following sections detail the per-channel transformations (adstock and saturation), the full revenue equation and confounding mechanism, and the target-ROI calibration.

2.2.1 Adstock (carryover)

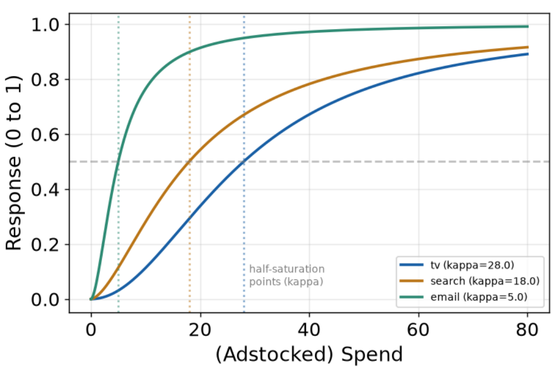
When we spend on advertising this week, the return effect doesn't all land this week.Someone sees your Facebook ad on Monday, thinks about it, and buys two weeks later. The effect sticks around and fades.



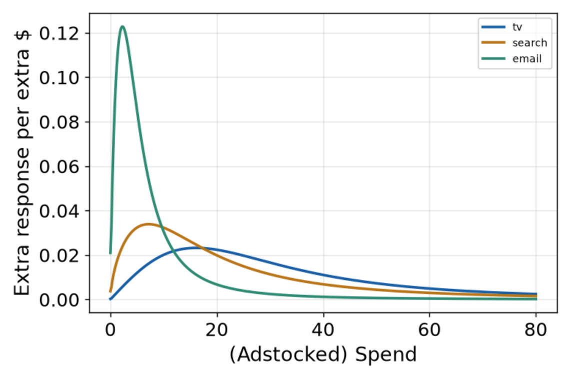
(a) Adstock Decay Profiles



(b) Geometric vs. Delayed Adstock



(c) Hill Saturation Curves



(d) Marginal Return Curves

Figure 2: Visualizing non-linear media transformations and channel decay behaviors. Panel **(a)** compares the learned adstock decay rates across channels, showing how long ad effects linger over time (e.g., TV has a longer memory than Search). Panel **(b)** contrasts standard geometric decay with a delayed-peak adstock formulation where the maximum impact is deferred. Panel **(c)** maps the Hill saturation curves to capture diminishing returns, highlighting each channel's half-saturation point (κ). Panel **(d)** illustrates the corresponding marginal returns (extra response per additional dollar spent), showing how marketing efficiency drops as cumulative spend increases.

Adstock is the transformation that captures this carryover effect. It is important to mention that this carryover effect varies depending on the marketing strategy we select (see fig. 2b). For example,

- **TV:** Long memory. Brand advertising lingers for weeks.
- **Search:** Short memory. Someone Googles, clicks, and buys; almost no carryover.

To model this carryover effect, we use the adstock function [1] below:

$$\text{adstock}(x_{t-L+1,m}, \ldots, x_{t,m}; w_m, L) = \frac{\sum_{l=0}^{L-1} w_m(l) x_{t-l,m}}{\sum_{l=0}^{L-1} w_m(l)} \tag{1}$$

This function transforms the time series of media spend in one channel. Here, L represents the maximum duration of the carryover effect, $x_{t,m}$ represents the media spend for channel m at time t, and w_m is a nonnegative weight function specific to the media channel. The weight function assigns importance to the media spend at different time lags, allowing for the modeling of the delayed impact of advertising.

The weight function w_m can be written in a number of different ways. The most common way is Geometric decay function [9, 1]. We denote that as:

$$w_m^g(l; \alpha_m) = \alpha_m^l, \quad l = 0, \ldots, L-1, \quad 0 < \alpha_m < 1, \tag{2}$$

where α_m is the retention rate of ad effect of the m-th media from one period to the next. This adstock function with geometric decay assumes that the advertising effect peaks at the same period as the ad exposure. But realistically, some media take a longer time to build such an effect. Therefore, peak effect may not happen at the same period as the ad exposure. To account for this delay we can use a delayed adstock function w_m^d as[1]:

$$w_m^d(l; \alpha_m, \theta_m) = \alpha_m^{(l-\theta_m)^2}, \quad l = 0, \ldots, L-1, \quad 0 < \alpha_m < 1, \quad 0 \le \theta_m \le L-1, \tag{3}$$

where θ_m is the delay of the peak effect. While this delayed formulation captures more complex lag structures, we restrict our analysis in this study to geometric decay to keep the modeling framework simple and highly interpretable.

2.2.2 Saturation (diminishing returns)

Doubling your marketing spend never doubles your results. The first marketing dollars are highly efficient because they reach most of the audience, those who haven't seen the ad. As we add more marketing spending, each additional dollar buys less incremental response. This is because we are targeting the same group of people too often. To model this diminishing-returns behavior, we use the Hill saturation function:

$$\text{Hill}_m(x_{t,m}; \mathcal{K}_m, \mathcal{S}_m) = \frac{x_{t,m}^{\mathcal{S}_m}}{\mathcal{K}_m^{\mathcal{S}_m} + x_{t,m}^{\mathcal{S}_m}}, \tag{4}$$

where $x_{t,m}$ is the adstocked media spend for channel m at time t, $\mathcal{S}_m$ is the shape parameter controlling how sharply the curve bends, and $\mathcal{K}_m$ is the half-saturation point: the spend level at which the response reaches half of its maximum value, i.e., $\text{Hill}_m(\mathcal{K}_m) = 0.5$. As $x_{t,m} \to 0$, the function approaches 0, it reflects no response at zero spend; as $x_{t,m} \to \infty$, the function approaches 1, it reflects full saturation.

Since the Hill function saturates at 1, we use scaling used by Yuxue et al. [1] to scale it by the channel's regression coefficient β_m so that each medium can have its own maximum achievable effect:

$$\beta_m \, \text{Hill}_m(x_{t,m}) = \beta_m \cdot \frac{x_{t,m}^{\mathcal{S}_m}}{\mathcal{K}_m^{\mathcal{S}_m} + x_{t,m}^{\mathcal{S}_m}}, \tag{5}$$

Here, β_m represents the maximum incremental effect the channel m can contribute once fully saturated: as $x_{t,m} \to \infty$, $\beta_m \, \text{Hill}_m(x_{t,m}) \to \beta_m$. Together, the adstock and saturation transforms define the complete media pipeline for each channel. Raw spend $x_{t,m}$ passes through two successive transformations before entering the revenue equation:

$$x_{t,m} \xrightarrow[\text{(carryover)}]{\text{adstock}} a_{t,m} \xrightarrow[\text{(diminishing returns)}]{\text{saturation}} s_{t,m} \xrightarrow{\times\,\beta_m} \text{contribution}_{t,m}. \tag{6}$$

The intermediate quantity $a_{t,m}$ (adstocked spend) is used by the Hill function and does not appear in our revenue model directly.

2.2.3 The revenue structural equation

In our study, we have five total media channels. Weekly revenue is generated by an additive structural model that combines a non-media baseline with the transformed contributions of the five media channels. Let $a_{t,m}$ denote the adstocked spend of channel m in week t (eq. (1)) and $s_{t,m}$ its saturated response (eq. (5)). The expected revenue in week t is

$$\mu_t \;=\; \underbrace{\beta_0}_{\text{baseline}} \;+\; \underbrace{\tau\, t}_{\text{trend}} \;+\; \underbrace{\sum_{k=1}^{K}\Big[\gamma_k \sin\!\big(\tfrac{2\pi k t}{P}\big) + \delta_k \cos\!\big(\tfrac{2\pi k t}{P}\big)\Big]}_{\text{yearly seasonality}} \;+\; \underbrace{\sum_{m=1}^{M}\beta_m\, s_{t,m}}_{\text{media contributions}} \;, \tag{7}$$

and observed revenue adds Gaussian noise,

$$y_t \;\sim\; \mathcal{N}\big(\mu_t,\; \sigma^2\big)\,. \tag{8}$$

Furthermore, yearly seasonality is represented by a Fourier basis with $K = 3$ sine–cosine pairs and period $P = 52.13$ weeks. We simulate $T = 156$ weeks (three years) with baseline $\beta_0 = 300$, trend slope $\tau = 0.6$, seasonal amplitude 40, and noise standard deviation $\sigma = 18$, all in thousands of dollars.

The eq. (7) has two identification-relevant assumptions. First, channel effects are additive (no interaction terms link one channel's effectiveness to another's activity). Second, all temporal dynamics of the media effect are carried by the adstock transform (conditional on $a_{t,m}$, past spend has no further influence on current revenue).

2.2.4 The confounding mechanism

We simulate two datasets: clean and confounded. These datasets use the exact same revenue equation and parameters. The only difference is that the confounded dataset adds a hidden demand variable z_t that influences both spend and revenue. This demand signal is built from a few components: a yearly seasonal cycle, a smaller six-month cycle, a slight upward trend, and random weekly noise,

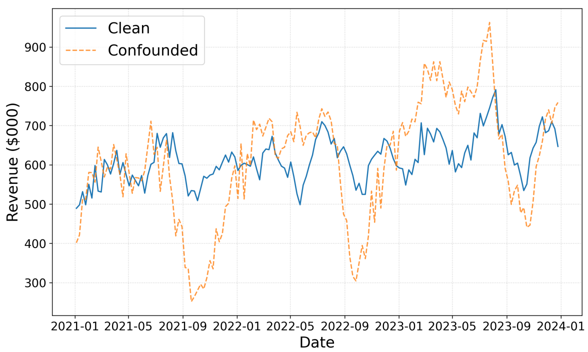
$$\tilde{z}_t \;=\; \sin\!\big(\tfrac{2\pi t}{P}\big) \;+\; 0.4\,\cos\!\big(\tfrac{4\pi t}{P}\big) \;+\; 0.01\,t \;+\; \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0,\, 0.3^2), \tag{9}$$

We standardize the raw signal as $z_t = (\tilde{z}_t - \bar{\tilde{z}})/\,\mathrm{sd}(\tilde{z})$, so that its effect size is controlled entirely by the scale parameters we name later. The random noise term ε_t is is a deliberate design choice: without it, demand would just be a predictable mix of trend and seasonality, and the model's baseline controls would absorb it completely. That would eliminate the confounding problem by luck rather than by proper adjustment, defeating the whole purpose of the experiment.

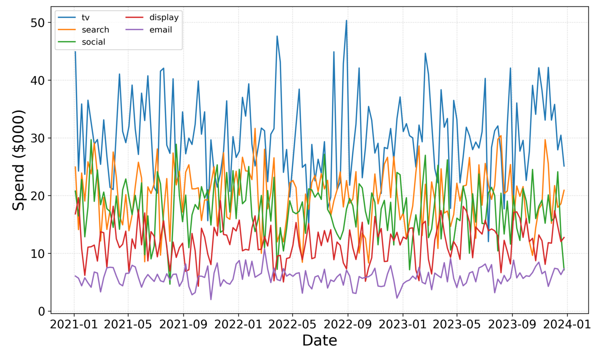
Demand enters the confounded dataset through two paths, matching the two arms of the backdoor path in Figure 1. First, it drives spend. Marketers increase budgets when they expect high demand. We model this multiplicatively,

$$x_{t,m} = \max\{\, \tilde{x}_{t,m} \cdot \exp(\lambda_m\, z_t)\,,\; 0.5\}\,, \qquad \tilde{x}_{t,m} \sim \mathcal{N}\big(p_m,\; v_m^2\big)\,, \tag{10}$$

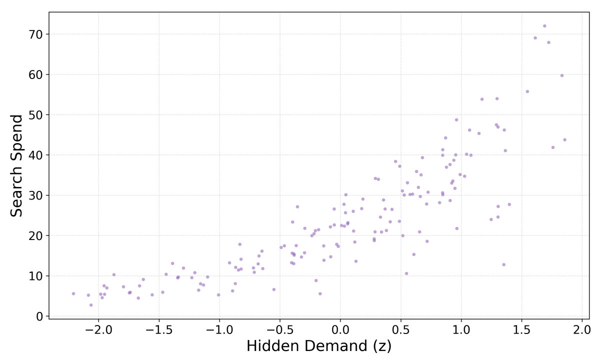
where $\lambda_m \geq 0$ controls how strongly each channel's spend reacts to the demand signal. The values range from $\lambda_{\text{email}} = 0.10$ to $\lambda_{\text{search}} = 0.60$, reflecting the real-world pattern that platform-driven channels like paid search react most aggressively to demand. The exponential form keeps spend positive and makes the response proportional: when demand is one standard deviation above average, search spend jumps by



(a) Clean vs. Confounded Revenue



(b) Weekly Channel Spend



(c) Confounding Relationship (Search)

Figure 3: Characterization of the synthetic dataset and the underlying confounding mechanism. Panel **(a)** illustrates how the latent demand signal z_t introduces highly volatile macro-environmental shocks into the confounded revenue series, contrasting with the stable, clean baseline. Panel **(b)** tracks the weekly budget allocations across the five marketing channels over the three-year simulation. Panel **(c)** reveals the root cause of the confounding bias: a strong positive correlation between Search spend and the unobserved demand variable z (where $\lambda_{\text{search}} = 0.60$), which opens the causal backdoor path.

roughly $e^{0.60} - 1 \approx 82\%$, while email spend barely moves at $e^{0.10} - 1 \approx 11\%$. Second, demand directly boosts revenue. We add a single term to eq. (7),

$$\mu_t^{\text{conf}} = \mu_t + \delta_z\, z_t, \qquad \delta_z = 45, \tag{11}$$

in thousands of dollars. To recover the clean dataset, we simply set $\lambda_m = 0$ for all channels and $\delta_z = 0$. Everything else stays the same. This means confounding is the single experimental knob we turn between the two datasets, and any difference in model performance between them is caused by the open backdoor path.

One important side effect of eq. (10): because all five channels respond to the same demand signal z_t, their spend series become correlated with each other (pairwise correlations as high as $r = 0.68$ between TV and search in the confounded data). This collinearity matters in the results because during estimation the confounder's influence gets spread across correlated channels. As a result, the bias in any one channel's ROI estimate doesn't simply track that channel's own λ_m, it depends on the correlations with other channels too.

2.2.5 The target-ROI calibration

For a synthetic validation study, we want to know the true ROI ahead of time, and it needs to be something we can interpret. We do that by choosing a target ROI for each channel and working backward to find the coefficient β_m.

While Zhang et al.[2] made their model for regional or city-level data, I adapted their method to work on a single national-level time series. By removing the regional layers, we can still use their mathematical approch to link the channel coefficient (β_m) directly to our target ROI_m, making it much easier to plug our experiment results straight into the model. Since ROI is just total contribution divided by total spend,

$$\text{ROI}_m = \frac{\sum_t \beta_m\, s_{t,m}}{\sum_t x_{t,m}} = \beta_m \cdot \frac{\sum_t s_{t,m}}{\sum_t x_{t,m}}, \tag{12}$$

we can solve for β_m:

$$\beta_m = \text{ROI}_m^{\text{target}} \cdot \frac{\sum_t x_{t,m}}{\sum_t s_{t,m}}, \tag{13}$$

where the sums are computed using the clean-world spend run through the true adstock and saturation transforms. We set target ROIs of $\{3.0, 4.5, 2.5, 1.8, 3.5\}$ for TV, search, social, display, and email, values that fall within the range seen in real-world MMM practice. The resulting β_m values are structural parameters: they represent what a dollar of advertising in each channel is truly worth, and they stay fixed across both the clean and confounded datasets.

However, the actual ROI in the confounded world ends up slightly different from these targets (for example, TV realizes 2.79 against a target of 3.0). This is because in a confounded world, spending gets concentrated in high-demand weeks, pushing more of each channel's dollars into the flat part of the Hill curve where returns are lower. To handle this, we score each experiment against its own world's realized ROI, clean-world targets for the recovery experiment, and confounded-world realizations for the confounding and calibration experiments. This way, any estimation error we measure reflects how well the model is doing, not a mismatch between baselines.

2.3 The three modeling approaches

We fit three models to clean data in an intentional order. We add complexity to each model that wasn't present in the previous model. Therefore, comparing these models tells us where the difficulty in estimating ROI actually lies. The first model ignores media transformations entirely. The second shows how well we could do if we already knew the transformations. In the third model, we don't know these transformations, and it has to learn them from the data.

2.3.1 Model 0: naive OLS

Model 0 regresses revenue on raw weekly spend, with no adstock and no saturation:

$$y_t \;=\; \beta_0 \;+\; \sum_{m=1}^{M} \beta_m \, x_{t,m} \;+\; \tau\, t \;+\; \text{seasonality}_t \;+\; \varepsilon_t. \tag{14}$$

Here, the baseline controls (linear trend and a three-pair Fourier seasonality basis) are the same as those used to generate the data. Therefore, any error in the recovered ROI is not caused by an incorrectly specified baseline, but rather by the media side of the model. Channel ROI is computed as total fitted contribution divided by total spend, $\widehat{\text{ROI}}_m = \sum_t \hat{\beta}_m x_{t,m} \big/ \sum_t x_{t,m}$, which for this model reduces to $\hat{\beta}_m$. The main purpose of model 0 is to quantify the cost of ignoring the two functional forms. Later, we include them because model 0's failure is the empirical justification for everything that follows.

2.3.2 Model 1: oracle OLS with known transformations

Model 1 applies the adstock and saturation transforms before running the same regression

$$y_t \;=\; \beta_0 \;+\; \sum_{m=1}^{M} \beta_m \, s^{\star}_{t,m} \;+\; \tau\, t \;+\; \text{seasonality}_t \;+\; \varepsilon_t, \tag{15}$$

where $s^{\star}_{t,m}$ is the saturated response computed with the *true* parameters $\theta^{\star}_m$, $\mathcal{S}^{\star}_m$ and $\mathcal{K}^{\star}_m$ taken directly from the data-generating process. Ordinary least squares then estimates only the scaling coefficients β_m. Realistically, this model is not achievable in practice. In any real application, the carryover rate and saturation curve of a channel are unknown.

Model 1 is therefore an oracle benchmark, not a method. Its purpose is to isolate a single question: if the functional forms were known exactly, how well would a simple regression recover ROI? When compared to model 0, the answer sets a maximum level of point-estimate accuracy and shows what the two transformations bring to the table.

2.3.3 Model 2: Bayesian MMM

This model is the realistic method and the centerpiece of our study. It uses the data to estimate the carryover rates, the saturation curves, the channel coefficients, the baseline, and the noise level. For each, it gives a full posterior distribution instead of a single number. We implement Model 2 in `PyMC-Marketing` (v0.19.3), sampling with the No-U-Turn Sampler. The generative model is the same as the structural model: geometric adstock, Hill saturation, a linear trend, a three-pair Fourier seasonality basis, and a Gaussian likelihood. What changes is that every parameter is now unknown and given a prior. The table table 1 lists them.

Table 1: Priors used in Model 2. All priors are specified on the internal scale of the estimator, which rescales spend and revenue to $[0, 1]$ using a maximum-absolute-value scaler.

Quantity	Our notation	Library name	Prior
Adstock retention	α_m	`adstock_alpha`	Beta(1, 3)
Saturation shape	$\mathcal{S}_m$	`saturation_slope`	Gamma(3, 1)
Half-saturation	$\mathcal{K}_m$	`saturation_kappa`	Beta(2, 2)
Channel coefficient	β_m	`saturation_beta`	HalfNormal(1)
Baseline	β_0	`intercept`	$\mathcal{N}(0, 2)$
Control coefficients	τ etc.	`gamma_control`	$\mathcal{N}(0, 2)$
Seasonality	γ_k, δ_k	`gamma_fourier`	Laplace(0, 1)
Observation noise	σ	`y_sigma`	HalfNormal(2)

We sample four chains of 1,000 draws each, following 2,000 tuning iterations, with a target acceptance probability of $0.98 - 0.99$ and a maximum tree depth of 13. A higher target acceptance is required because the Hill saturation parameters trade off against one another along a curved ridge in the posterior, a well-documented identification difficulty in media mix models. Convergence is assessed by the Gelman-Rubin statistic ($\hat{R}$), effective sample size, and divergent-transition counts (see section 3.4).

2.4 The calibration experiment

The backdoor adjustment described in Stage 2 of our experiment succeeds only because the confounding variable (demand) was documented in our synthetic data. In real-world applications, we don't have this confounding variable. When this confounding variable cannot be measured, it cannot be conditioned on, and backdoor adjustment is unavailable. This section describes the alternative approach considering Zhang et al. [2].

2.4.1 Lift testing in practice

To resolve this unmeasurable confounding, firms use experimentation, such as geo-lift tests. In these tests firms divide their markets into random treatment and control groups. Then pause one channels advertising for a fixed period of time. Finally, they compare the outcome of these two groups and ultimately calculate the experimentally measured ROI.

However, these lift tests are expensive, and these tests reduce the revenue in the controlled group. Additionally, these tests takes weeks to run because they usually address one channel at a time. Modern marketing firms cannot afford run all the experiments, and that is why they use observational MMM in their applications. However, usual MMM calculations are subject to confounding issues. To overcome that, we use the approach introduced by Zhang et al. [2], which combines MMM with lift tests.

2.4.2 Simulating the experiment

In this study, we will not conduct an actual lift test. As an alternative, we tried to get what we thought we would get from a lift test: an unbiased estimate of ROI that is close to the real value. These values are centered on the true value with a standard error that reflects the uncertainty you would get from a real experiment with a finite sample. For example, we adjust the search channel, which shows the strongest link to demand. Its true ROI in the confounded world is 3.83. We simulate a lift test returning

$$\widehat{\mathrm{ROI}}^{\mathrm{exp}}_{\mathrm{search}} \;=\; 3.9, \qquad \mathrm{SE} \;=\; 0.4, \tag{16}$$

we keep this small standard error intentionally because a real lift test would never return an absolute true value. Instead, it produces slightly closer estimates, just as in our approach. Since we know and use the true ROI, this experiment is an oracle, just like model 1. What the calibration experiment shows is how much value accurate experimental evidence adds when you have it.

2.4.3 Injecting the experiment as a prior

A prior is just what you believe about a parameter before looking at your current data. Rather than guessing, we are using ROI estimate from the above independent randomized experiment(simulated lift test in our case). However, our experiment gives us an answer in ROI units, but the model works with the scaled coefficient β_{search}. Therefore, we need to convert between the two. On the model's internal scale, a channel's total contribution in original units is $y_{\max}\,\beta_m \sum_t s_{t,m}$, where $y_{\max}$ is the scaling constant applied to revenue. Setting this equal to $\mathrm{ROI}_m \sum_t x_{t,m}$ and solving gives

$$\beta_m \;=\; \mathrm{ROI}_m \cdot \frac{\sum_t x_{t,m}}{y_{\max} \sum_t s_{t,m}}. \tag{17}$$

When we apply above equation to the simulated lift-test estimate and its standard error yields a prior of $\mathcal{N}(0.179,\, 0.018^2)$, truncated below at zero to preserve the sign constraint. As a consistency check, we inverted the conversion and confirmed that we recover the original value exactly: we have substituted $\beta = 0.179$ back into the forward calculation, recovering an implied ROI of 3.9, matching the injected estimate exactly. For reference, the value of β_{search} corresponding to the *true* ROI of 3.83 is 0.176, so the prior is centered slightly above the truth, as intended. The calibrated model (model 3) is the same as model 2 with one prior substituted, fitted on the confounded data with the demand variable intentionally removed from the control. The backdoor path therefore remains open throughout.

2.4.4 Adjustment versus calibration

Two solutions we examine in this study deal with the same confounder in fundamentally different ways.

- **Backdoor adjustment** blocks the confounding path. It requires you to actually measure the confounder and include it in the model. By conditioning on demand directly, the spurious link between spend and revenue is removed before the causal effect is estimated.
- **Experimental calibration** bypasses the confounding path entirely. It does not need to measure confounders. Instead, it gets a clean estimate of the causal effect from a randomized experiment that is already immune to confounding by design and uses that estimate to correct the biased observational data.

When we know the confounding variable, backdoor adjustment is the cheap option. But when we don;t know the confounding variables, MMM with calibration is the best approach, especially in the production setting.

3 Results

The central question of this study is whether an observational MMM can be trusted to recover causal ROI, and if not, what it takes to make it trustworthy. We do not propose new estimation methodology; the Bayesian MMM, the backdoor-adjustment framework, and the calibration-via-prior technique are all drawn directly from existing work [1, 2, 3]. Our contribution is empirical demonstration: using synthetic data with known ground truth, we show concretely when these established methods succeed, when they fail, and what it takes to correct them. We organize this demonstration in four progressive stages: (1) baseline model validation on clean data, (2) the isolation of observational bias on confounded data, (3) the application of experimental calibration, and (4) the replication of these phenomena using an applied real-world dataset.

3.1 Recovery on clean data

Before we do any demonstration about confounding, we have to show that these methods work in the most favorable possible setting: clean data and media spend assigned independently of everything else, so the backdoor path of fig. 1 does not exist. If our model cannot pass this basic test on clean data, we will not know whether its future failures are caused by confounding or simply a flawed model.

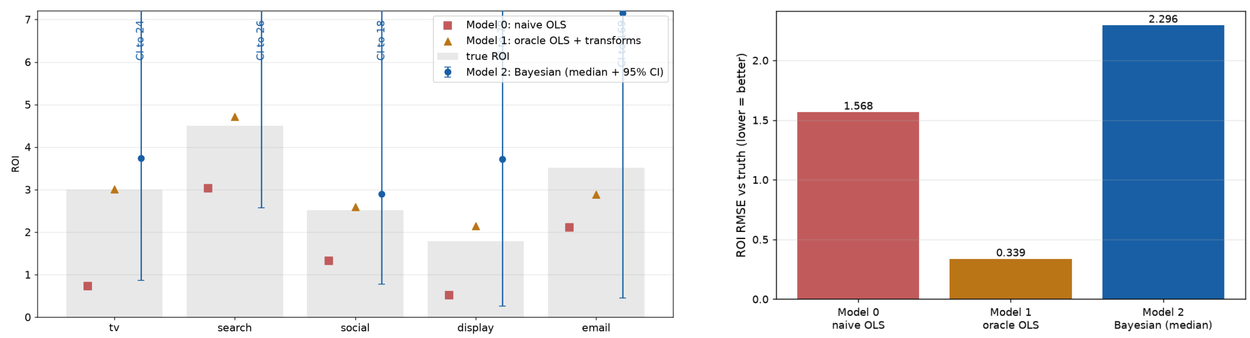
Table 2: Comparison of True ROI Estimates Across All Models on Clean Synthetic Data

Channel	True ROI	Model 0	Model 1	Model 2 (Median)	Model 2 (95% CI)
TV	3.00	0.733	3.011	3.74	[0.87, 24.16]
Search	4.50	3.029	4.719	7.44	[2.58, 25.60]
Social	2.52	1.328	2.590	2.90	[0.78, 18.00]
Display	1.79	0.518	2.142	3.72	[0.26, 79.31]
Email	3.51	2.108	2.882	7.15	[0.45, 169.09]

Table 3: Model Fit and Error Comparison for Naive and Oracle OLS

Metric	Model 0	Model 1
RMSE	1.568	0.339
R^2	0.888	0.916

Transformations Drive Model Accuracy. When we compare our baseline models, results show us why standard modeling fails (table 2). The naive OLS model (model 0), which uses raw ad spend, significantly underestimates ROI. It reports 0.73 instead of 3.00 for TV and 0.52 instead of 1.79 for Display. This underestimation happens mainly the model 0 ignores adstock and saturation effects. Simply applying these correct mathematical transformations (model 1/oracle OLS) eliminates almost all of this bias. For example, estimated TV ROI jumps to 3.01, Search rises to 4.72, and overall RMSE drops from 1.57 to 0.34 (with R^2 increasing from 0.888 to 0.916). These results demonstrate that for a clear ROI estimation, modeling the media response shape matters far more than the choice of estimator itself.



(a) Per-Channel ROI Estimates vs. Truth

(b) Aggregate ROI RMSE Comparison

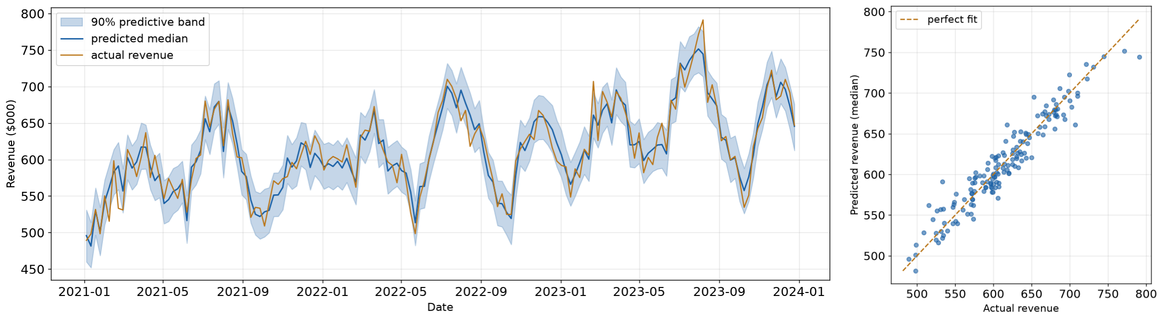
Figure 4: Comparison of baseline model estimates against ground-truth ROI under data confounding. Panel **(a)** displays the point estimates and uncertainty intervals for each channel. While the uncalibrated Bayesian model (Model 2) successfully captures the true ROI within its 95% credible intervals, the intervals are extremely wide (extending up to 69 for Email) and the median point estimates overshoot wildly due to weak parameter identification. Panel **(b)** quantifies this estimation error using aggregate ROI RMSE. Because the uncalibrated Bayesian medians drift so far from the truth, Model 2 produces a significantly higher overall error (2.296) than even the naive OLS model (1.568), demonstrating why experimental calibration is mathematically required to anchor the posteriors.

Bayesian Parameter Recovery. We have used real transformations from the data generation in model 1, but for actual cases, we need to learn these transformations directly from the data as full posterior distributions instead of single point estimates. Therefore, we employ model 2, and it learns these transformations from the data using Bayesian modeling. In model 2 results (table 2), every channel's true ROI falls within its 95% credible interval. The model also shows a strong posterior-predictive fit, placing 94.2% of observed weeks within the 90% predictive interval and achieving a predictive R^2 of 0.91 (figs. 5a and 5b).

However, we must admit that the credible intervals for channels like Email and Display are extremely wide. This happens because these channels have low spend with little variation(see fig. 3b), so the data has little to no information about where their saturation curves bend, the model can fit the observed revenue equally well under many different ROI values. This demonstrates that while the uncalibrated Bayesian model successfully covers the true ROI, the resulting estimates are often too imprecise to guide practical business decisions. Narrowing these intervals requires either channel-level experimental evidence or spend variation the historical data does not contain.

3.2 Experimenting on confounded data

Next, we introduce confounding by turning on the latent demand signal z_t, which affects both ad spend and sales. Since we keep the model structure and priors exactly the same, any changes in the calculated ROI are caused by this confounder alone. We run this same Bayesian MMM model twice: once leaving the demand confounder uncontrolled (naive), and once conditioning on demand (adjusted).



(a) Time-Series Predictive Coverage (b) Predicted vs. Actual

Figure 5: Posterior predictive checks (PPC) for the Bayesian model (Model 2) evaluated on the unconfounded (clean) dataset. Panel **(a)** tracks the 90% predictive interval band and the predicted median against the actual clean revenue time series. Panel **(b)** compares the predicted median revenue against the true revenue along the 45-degree line of perfect fit. The exceptionally tight tracking and balanced error distribution validate the model's structural specification; in the absence of confounding, the model successfully learns the true underlying parameters of the data-generating process.

Table 4: Confounding Bias on ROI: Naive vs. Adjusted Model

Channel	**True ROI**	**Naive ROI**	**Naive Bias**	**Adjusted ROI**	**Adjusted Bias**	**Demand β**
TV	2.79	2.26	-0.53	2.32	-0.48	0.45
Search	3.83	6.57	+2.74	4.81	+0.97	0.60
Social	2.41	2.50	+0.09	1.79	-0.62	0.30
Display	1.78	0.88	-0.91	0.76	-1.02	0.20
Email	3.50	11.49	+7.99	9.22	+5.72	0.10

Table 5: Aggregate Media Contribution (Over 3 Years)

Model Context	**Contribution ($)**	**Error vs. Truth**
True Total (from DGP)	41,450	–
Naive Posterior Median	63,931	+54.2%
Adjusted Posterior Median	59,479	+43.5%

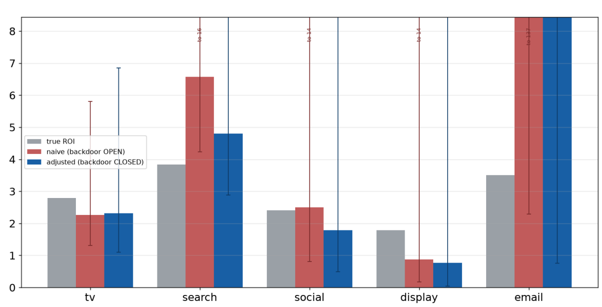
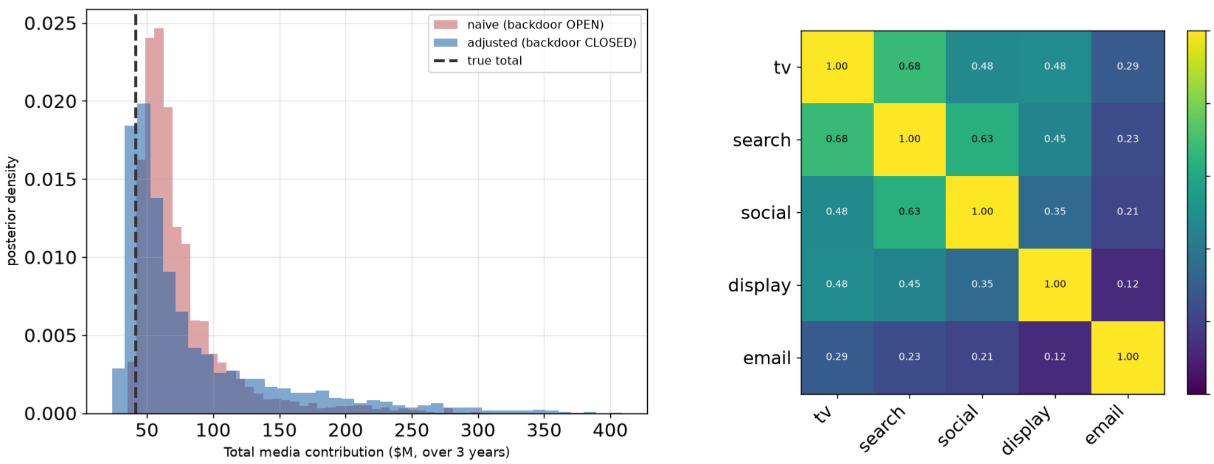


Figure 6: Per-channel ROI estimation errors under data confounding across different model specifications. The gray bars represent the true realized ROI in the confounded world (e.g., 3.83 for Search). The red bars show the estimates from the naive model where the causal backdoor path is left open; notice how the model drastically overestimates Search performance (> 6.5) because it attributes organic demand-driven sales to ad spend. The blue bars demonstrate the impact of closing the backdoor path using experimental calibration; the adjusted model successfully mitigates this confounding bias, pulling the median estimates back toward the true baseline.

Confounding Bias Affects Channels Unevenly. Leaving the demand confounder uncontrolled (backdoor open) does not bias all channels equally. It heavily inflates Search (+2.74) and Email (+7.99) while leaving other channels mostly unaffected (table 4). Given the strong correlation between search and demand ($\lambda_{\text{search}} = 0.60$), it makes sense that its estimated ROI has gone up. Surprisingly, Email suffers the worst bias despite having the weakest correlation to demand ($\lambda_{\text{email}} = 0.10$). This shows that individual channel bias does not directly match demand correlation.

This mismatch happens for two reasons. First, because all five channels rise and fall with the same demand signal z_t, their spend levels are highly correlated (fig. 7b). The regression model struggles with this correlation. Second, Email's inflation is not entirely caused by the confounder. When we check the clean dataset results, Email's ROI was already highly overestimated (median 7.15 vs. true 3.51). Therefore, most of the Email's bias in the confounded model is actually just this pre-existing estimation error carrying over, it is not the confounder. This difference is important because adding a demand control variable can fix confounding (backdoor closing), but it cannot fix a model that is already weak identifying a specific channel.



(a) Total Media Contribution Posteriors

(b) Spend Correlation Matrix

Figure 7: Confounding signatures in aggregate media impact and spend dynamics. Panel **(a)** shows that the naive model (red) heavily overestimates the total media contribution because of the open backdoor path, whereas the adjusted model (blue) successfully centers around the true total (dashed line). Panel **(b)** displays the correlation matrix of weekly spend across the five channels in the confounded dataset. The strong positive correlations (such as TV, Search, and Social) reflect how their budgets are mutually dragged upward by the unobserved demand signal.

Adjustment Only Helps Certain Channels. Controlling for demand closes the backdoor path, but only for channels where the bias actually comes from that path. Search is the best example: its bias falls from +2.74 to +0.97. However, this adjustment does not fix everything. Email remains highly inflated even after controlling for demand (+5.72, down from +7.99). This confirms our earlier point: adding a demand control cannot fix a structural identification problem that occurs even with clean data. Looking at the big picture, the total three-year media contribution is overestimated by +54.2% in the naive model, and still by +43.5% after adjustment (table 5 and fig. 7a). Finally, from the stage 2 experiments, we can state that backdoor adjustment fixes strongly confounded channels that the model can easily identify, but it leaves behind bias when confounding is mixed with high channel correlation or weak identification.

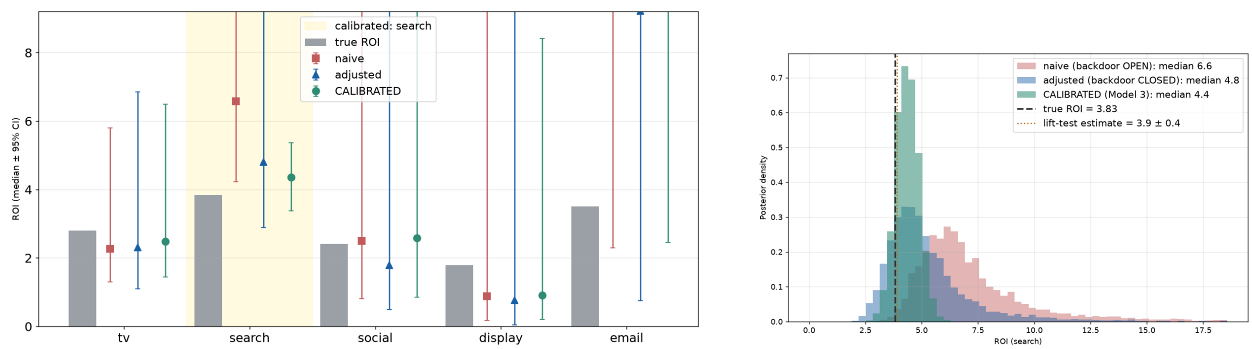
3.3 Experimental calibration

The adjustment of section 3.2 worked only because z_t was observed when we were constructing the data. In real-life cases, it is difficult or nearly impossible to observe this confounding variable. In the third stage, we experiment with an alternative approach [2]. We don't use the confounding variable (leave the backdoor open), but we replace the weakly informative prior on one channel's coefficient with an informative prior derived from a simulated geo-lift test. We select search as the one channel because it has the highest demand correlation. For the search, we did a simulated lift test with a list estimate of $\widehat{\text{ROI}} = 3.9$ with $\text{SE} = 0.4$ (see section 2.4.2).

Table 6: Calibration Summary: Naive, Adjusted, and Calibrated ROI (Lift Test on Search: 3.9 ± 0.4)

Channel	**True ROI**	**Naive ROI**	**Adjusted ROI**	**Calib. Median**	**Calibrated 95% CI**
TV	2.79	2.26	2.32	2.49	[1.44, 6.49]
Search	3.83	6.57	4.81	4.35	[3.37, 5.36]
Social	2.41	2.50	1.79	2.59	[0.85, 15.65]
Display	1.78	0.88	0.76	0.91	[0.20, 8.41]
Email	3.50	11.49	9.22	11.66	[2.45, 108.69]

Calibration Anchors the Target Channel. When we calibrate the search channel, the estimated ROI shrinks toward the true value (table 6), the naive estimate (6.57) and the adjusted estimate (4.81) down to 4.35. More importantly, the range of uncertainty tightens from a wide naive interval to a compact $[3.37, 5.36]$ (see fig. 8b). This experiment does this improvement with the backdoor open. Even though the observational data are too tangled, the experimental prior provides the exact information the model needs to separate cause from effect. These results are a clear demonstration that a small piece of randomized evidence can correct a bias that observational data alone could never fix.



(a) Per-Channel Calibration Impact

(b) Search ROI Posterior Distributions

Figure 8: The impact of single-channel experimental calibration on ROI estimation under confounding. Panel **(a)** provides an overview across all five media channels, highlighting the Search channel (yellow band) where the experimental prior is applied. Incorporating the lift-test constraint drastically reduces uncertainty and anchors the calibrated estimate closer to the true baseline compared to the uncalibrated models. Panel **(b)** isolates the Search channel's posterior density shifts. Leaving the causal backdoor open (naive, red) forces the model to settle at an inflated median of 6.6. Closing the backdoor via control adjustments (blue) drops the median to 4.8 but leaves a wide variance. Introducing the lift-test prior (3.9 ± 0.4, dotted line) yields the calibrated model (green), which successfully shrinks posterior variance and pulls the median down to a highly accurate 4.4, tightly mirroring the true ground-truth ROI of 3.83 (dashed line).

However, single calibration does not fix the whole model. For example, the uncalibrated Email channel remains highly inflated at a median ROI of 11.66, it is worse than the adjusted ROI. Therefore, calibration only fixes the targeted channel and slightly stabilizes highly correlated channels; it does not rescue the entire model. This limitation is exactly why we argue that calibration is not a one-time fix but a continuous process that must be repeated wherever experiments are actually run.

3.4 Application to realistic data (Robyn)

The synthetic stages establish the mechanism in a world where we control the truth. In the stage 4, we check for the same patterns in data we did not build. We fit the Bayesian MMM to Meta's open-source Robyn dataset (208 weeks, five paid channels) under two specifications: a well-specified model that adjusts for market demand through a competitive-sales control (closed backdoor), and a counterfactual model that omits it (open backdoor).

Channel	Well-Specified ROI	Omitted Confounder ROI	Absolute Shift	% Change
TV	7.59	6.35	−1.24	−16.4%
OOH	0.11	0.12	+0.01	+11.9%
Print	5.76	4.47	−1.29	−22.4%
Facebook	13.79	18.29	+4.49	+32.6%
Search	5.39	9.34	+3.95	+73.3%

Table 7: Observational bias in applied MMM (Robyn Dataset). The table compares posterior median ROIs from a well-specified model (controlling for competitor sales) against a counterfactual model where the primary market demand confounder is omitted. Lower-funnel digital channels (Search, Facebook) absorb the unobserved demand, severely inflating their estimated ROI.

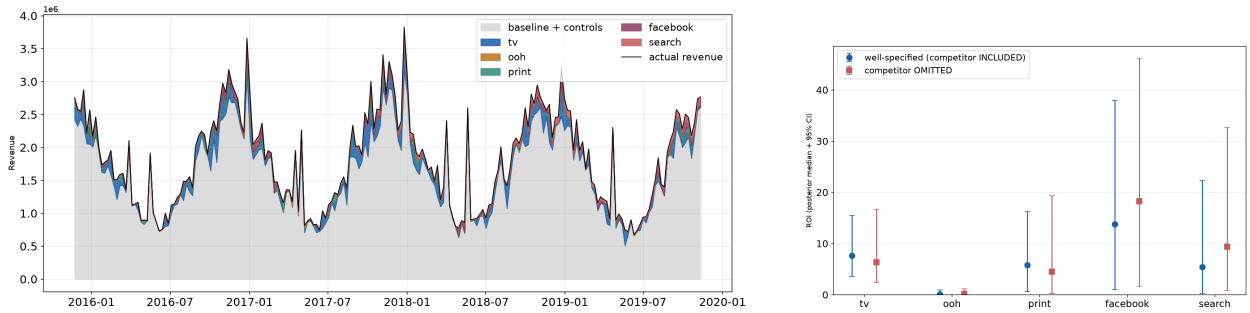
Channel	Spend ($M)	Contrib.	Median ROI	95% CI
TV	3.09	6.1%	7.59	[3.59, 15.44]
OOH	8.99	0.2%	0.11	[0.00, 0.93]
Print	0.78	1.1%	5.76	[0.64, 16.19]
Facebook	0.45	1.5%	13.79	[0.98, 38.00]
Search	1.23	1.6%	5.39	[0.27, 22.27]

Table 8: Applied summary of the well-specified Bayesian MMM fit using the Robyn dataset. The table displays total channel spend in millions, overall media contribution percentage, and the posterior median Return on Investment (ROI) alongside 95% credible intervals.

Real Data Results Mirror Synthetic Failure. When we drop the competitor-sales control, results shows the same pattern shown in the synthetic experiment (see table 7). In the absence of this crucial control, the unobserved market demand is absorbed by lower-funnel digital channels, leading to a significant increase in ROIs (see table 8). For example, Search's median ROI rises by +73.3% (5.39 → 9.34) and Facebook's by +32.6% (13.79 → 18.29). On the other hand, the upper-funnel and offline channels move little or in the opposite direction (TV −16.4%, print −22.4%).

Consistency, Not Proof. Because the Robyn data has no known ground-truth ROI, we cannot verify that the well-specified model is correct. It simply confirms that omitting a demand variable shifts estimates in the exact direction and pattern shown in our synthetic experiment. Computationally, both specifications encountered a high number of divergent transitions under NUTS, with the well-specified model sampling notably slower. This is because when we include a strong market indicator (competitor spend), it causes multicollinearity.

Summary of the Four Stages. Stage 1 demonstrates that a well-specified Bayesian MMM successfully recovers causal ROI under proper identification. Stage 2 shows how an unmeasured confounder breaks this recovery, unpredictably redistributing bias across correlated channels in a way that standard controls can only partially repair. Stage 3 proves that injecting a single randomized lift estimate as a prior corrects the calibrated channel where observational data alone fails. Finally, Stage 4 replicates this exact confounding signal in real-world data, confirming the underlying mechanism is not a synthetic artifact. We therefore



(a) Historical Revenue Decomposition

(b) Empirical Omitted Variable Bias

Figure 9: Empirical validation of confounding and omitted variable bias on a real-world multi-channel dataset. Panel **(a)** displays the historical time-series revenue decomposition across media streams and baseline controls, showing tight alignment with actual sales. Panel **(b)** serves as the real-data validation of our synthetic confounding theory by observing what happens when the strongest market control variable competitor marketing activity is intentionally dropped from the model. Omitting competitor spend (red) severely disrupts parameter identification, shifting the median ROI point estimates for high funnel digital channels like Facebook and Search while triggering a massive inflation in their 95% credible intervals.

conclude that navigating real-world deployment requires a unified approach, combining observational MMM with randomized lift testing to achieve optimal, causally robust performance.

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Appendix

Table 9: Posterior Summary Statistics of Model 2

Parameter	**Mean**	**SD**	$\hat{R}$	**ESS (Bulk)**
Intercept	-0.233	0.529	1.002	1046.06
Adstock Retention (α)				
TV Spend	0.575	0.096	1.001	2240.78
Search Spend	0.137	0.071	1.001	2688.46
Social Spend	0.288	0.143	1.001	2156.26
Display Spend	0.348	0.216	1.001	2993.14
Email Spend	0.317	0.196	1.001	3143.12
Saturation Slope				
TV Spend	2.198	1.378	1.000	1913.45
Search Spend	1.220	0.647	1.000	2122.85
Social Spend	2.135	1.478	1.003	1951.06
Display Spend	2.655	1.711	1.000	2140.17
Email Spend	2.570	1.649	1.001	2281.16
Saturation κ (Shape)				
TV Spend	0.519	0.222	1.001	2003.05
Search Spend	0.524	0.212	1.001	3218.93
Social Spend	0.498	0.205	1.003	2450.13
Display Spend	0.385	0.251	1.001	1380.37
Email Spend	0.380	0.253	1.002	1464.92
Saturation β (Scale)				
TV Spend	0.350	0.279	1.001	1522.40
Search Spend	0.413	0.256	1.000	1912.76
Social Spend	0.167	0.159	1.002	1482.85
Display Spend	0.220	0.325	1.001	1213.26
Email Spend	0.219	0.326	1.002	1385.62
Observation Noise (σ)	0.023	0.001	1.001	4234.24